ADUKUNDHAM

Family Fun & Games Hub

Project Documentation

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1. Introduction

Adukundham is a web-based entertainment platform designed to provide simple, interactive, and enjoyable games and activities for users.

The application brings multiple games and entertainment activities together in a single website. Users can access different games directly from the platform without needing separate applications.

The main purpose of Adukundham is to provide a simple and user-friendly environment where friends and family members can spend time together through interactive games and quizzes.

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2. Problem Statement

Many casual games and entertainment activities are available across different websites and applications. Users often need to search for different platforms to access different games.

Adukundham addresses this problem by providing multiple interactive activities in one centralized web application.

The application focuses on:

- Easy access to games
- Simple user interface
- Browser-based usage
- Family and friends entertainment
- Multiple activities in a single platform

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3. Objectives

The main objectives of Adukundham are:

- To develop a simple and attractive entertainment website.
- To provide multiple games in one platform.
- To make the application easy to use for different age groups.
- To provide browser-based games without requiring installation.
- To create an interactive experience for friends and family.
- To provide a foundation for adding more games and features in the future.

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4. Main Features

4.1 Multiple Games

The application provides multiple games and activities through a single website.

Current modules include:

- Tic-Tac-Toe
- Sudoku
- Quiz
- Cards

Each activity is implemented as a separate module and can be accessed from the application.

4.2 Simple User Interface

The application is designed with a straightforward interface so that users can easily navigate between different activities.

4.3 Browser Based

Adukundham can be accessed through a modern web browser. Users do not need to install a separate desktop application.

4.4 Interactive Gameplay

The games provide interactive elements that allow users to actively participate rather than simply viewing content.

4.5 Modular Structure

Each game is maintained as a separate HTML page/module. This makes the project easier to maintain and allows additional games to be added later.

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5. Application Modules

5.1 Tic-Tac-Toe

Tic-Tac-Toe is a two-player strategy game played on a 3 × 3 grid.

Players take turns placing their respective symbols on the board. The player who successfully places three symbols in a row, column, or diagonal wins the game.

5.2 Sudoku

Sudoku is a number-based logic puzzle.

The objective is to fill the Sudoku grid so that every row, column, and required sub-grid contains the appropriate numbers without repetition.

The Sudoku module provides users with an interactive puzzle-solving experience.

5.3 Quiz

The Quiz module provides interactive questions that allow users to test their knowledge.

A quiz system can include:

- Multiple-choice questions
- Answer selection
- Score calculation
- Correct/incorrect feedback
- Final score display

5.4 Cards

The Cards module provides a card-based entertainment activity.

It can be extended in the future with different card games and additional gameplay functionality.

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6. Technologies Used

The current project is implemented using standard web technologies.

HTML5

HTML is used to create the structure and content of the web pages.

CSS3

CSS is used for:

- Layout
- Colors
- Typography
- Buttons
- Game boards
- Responsive design
- Visual styling

JavaScript

JavaScript is used to provide interactive functionality and game logic.

It can handle:

- User interactions
- Game rules
- Score calculation
- Dynamic page updates
- Event handling

GitHub

GitHub is used for source-code hosting and project version control.

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7. Project File Structure

The project currently contains separate files for different application modules.

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├— index.html

├— cards.html

├── quiz.html

├── sudoku.html

├── tictactoe.html

├── style.css

├── script.js

└── README.md

File Description

index.html
Main/home page of the application.

cards.html
Cards-related activity.

quiz.html
Interactive quiz module.

sudoku.html
Sudoku game module.

tictactoe.html
Tic-Tac-Toe game module.

style.css
Contains the styling and visual design of the website.

script.js
Contains JavaScript functionality and interactive logic.

README.md
Contains project information and documentation.

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8. How the Application Works

The application follows a simple web-based architecture.

User

↓

Home Page

↓

Select Activity

↓

Game / Quiz Module

↓

User Interaction

↓

JavaScript Processing

↓

Result / Score / Game State

The user first enters the application through the home page.

From there, the user can select an available activity. The corresponding HTML page is opened and JavaScript manages the interactive functionality.

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9. System Requirements

Hardware Requirements

- Computer, laptop, tablet, or smartphone
- Minimum 2 GB RAM recommended
- Internet connection for accessing the hosted website

Software Requirements

- Modern web browser
- Google Chrome
- Microsoft Edge
- Mozilla Firefox
- Safari

No additional software installation is required for normal browser usage.

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10. Advantages

Adukundham provides several advantages:

- Multiple activities in one platform.
- Easy-to-use interface.
- Browser-based access.
- Simple project architecture.
- Easy maintenance.
- Easy addition of new games.
- Suitable for casual entertainment.
- Can be accessed from different devices.

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11. Future Enhancements

The application can be extended with several additional features.

User Accounts

Users could create accounts and maintain their profiles.

Leaderboards

A leaderboard could be introduced to display the highest scores of users.

Multiplayer Games

Real-time multiplayer functionality could allow users to play games with friends remotely.

More Games

Future versions could include:

- Chess
- Checkers
- Memory games
- Word games
- Snake
- 2048
- Rock Paper Scissors
- Connect Four

Progressive Web App

The website could be converted into a Progressive Web App (PWA), allowing users to install it on supported devices.

Database Integration

A database could be added to store:

- User profiles
- Scores
- Game history
- Leaderboards
- Achievements

Mobile Application

A dedicated Android/iOS application could be developed using technologies such as Flutter.

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12. Project Scope

Adukundham can be developed beyond a simple collection of games into a complete digital family entertainment platform.

The long-term vision is to provide users with a centralized platform containing games, quizzes, challenges, leaderboards, and social interaction features.

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13. Conclusion

Adukundham is a web-based family entertainment platform that combines multiple interactive games and activities into one application.

The project demonstrates the practical use of HTML, CSS, and JavaScript to create interactive web applications.

Its modular structure makes it easy to maintain and provides a strong foundation for adding more games, multiplayer functionality, user accounts, leaderboards, and other advanced features in future versions.

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14. Developer Information

Project: Adukundham
Category: Web Application / Entertainment Platform

Developer: Saipavan Blip

Repository:
GitHub – Adukundham

The project is maintained as an open-source/public repository and can be further developed with additional games and functionality.

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15. Project Summary

Category	Details
Project Name	Adukundham
Type	Web Application
Domain	Entertainment / Games
Frontend	HTML, CSS, JavaScript
Repository	GitHub
Main Modules	Cards, Quiz, Sudoku, Tic-Tac-Toe
Platform	Web Browser
Target Users	Friends and Family
Future Scope	Multiplayer, Accounts, Leaderboards, More Games

Final Statement

Adukundham – Play. Challenge. Enjoy.

A simple platform bringing games and entertainment together in one place.